



EE712 - Embedded Systems Design

Electrical Engineering IIT Bombay

Laboratory Experiment - III

Problem set: 3

Date: January 30, 2025

LCD Interfacing and On-Chip Temperature Sensor

Problem 1

Aim: To get familiarized with the 16 X 2 LCD display and interface the LCD display with TIVA C Board.

Components to be used:

- TIVA C Board
- 16 X 2 LCD Display

Problem Statement:

- Given the snippet of a LCD interfacing code lcd.c.
- Complete all the 4 tasks given inside the snippet. You can refer to the following pdf for more information about LCD. LCD_Control_Made_Easy
- Add the header files as required in tasks 1 and 2 in the project by right-clicking on the project – > New files – > Header file.
- After completing the tasks, flash the code into TIVA board and print first 3 letters of your teams member's name in the first row and ESD - IIT Bombay in the 2nd row.
- *E.g. If the member names are Aksh, Tejas, Ritam and Tejaswee, display on LCD:*

AKS-TEJ-RIT-TEJ
ESD-IIT Bombay

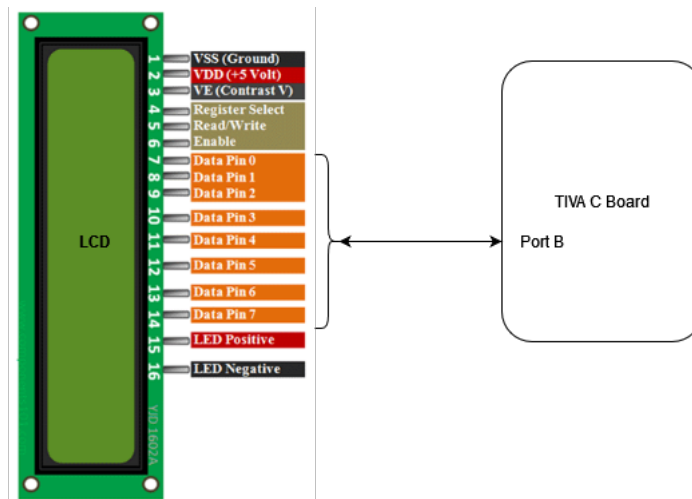


Figure 1: LCD connection to TIVA-C board

Problem 2

Aim: To measure the temperature of the room using the on-chip temperature sensor and on-chip ADC and display it on the LCD.

Components to be used:

- TIVA C Board
- 16 X 2 LCD Display

Problem Statement:

- Make appropriate changes to your code that you have written for the **LAB 2 - problem set 3 : ADC interfacing** so that you can read the value from the on-chip temperature sensor.
- Calculate the temperature reading (in °C) from the internal sensor using the formula:
$$TEMP = 147.5 - ((75 * (VREFP - VREFN) * ADCValue) / 4096)$$
- Refer TM4C123GH6PM Microcontroller datasheet, Page 812 for more details.
- Display the obtained value upto 2 decimal values on LCD display.

General Instructions

- Include the following header files in your code:
"stdint.h", "stdbool.h", "inc/hw memmap.h", "inc/hw types.h", "driverlib/gpio.h", "driverlib/sysctl.h", "inc/hw ints.h", "driverlib/interrupt.h", "driverlib/adc.h".
- Enable the system clock, ADC0 peripheral clock and the required port pins.
- Check the TM4C123GH6PM datasheet, Page 801 to find which pin can be configured as ADC input. Use GPIOPinTypeADC(), Page 266 to set.
- Set the enabled port pins to function as output pins.
- Configure the trigger source for ADC and priority of a sample sequence using the function ADCSequenceConfigure(), Page 38.
- Configure the ADC channel using the function ADCSequenceStepConfigure(), Page 42.
- Enable the sample sequence using the function ADCSequenceEnable(), Page 41.
- Trigger a sample sequence with processor using the function ADCProcessorTrigger(), Page 36.
- Wait until the sample sequence has completed by checking ADCIntStatus(), Page 34.
- Read the digital value from the ADC using the function ADCSequenceDataGet(), Page 39.
- Triggering should be done after every sample conversion.
- Calculate the temperature reading (in °C) from the internal sensor using the formula:
$$TEMP = 147.5 - ((75 * (VREFP - VREFN) * ADCValue) / 4096)$$
- Refer TM4C123GH6PM Microcontroller datasheet, Page 812 for more details.
- Display the obtained digital value in decimal format on LCD screen.

Console configuration:

Go through **Page 5-13 to 5-15** to configure the variable to display it in the CCS console of the following link.
TM4C123G_LaunchPad_Workshop_Workbook

References:

1. TM4C123GH6PM : TM4C123GH6PM_Datasheet
2. TIVA Driver Library: TIVA_Driver_Library
3. TM4C123G LaunchPad Workshop Workbook TM4C123G_LaunchPad_Workshop_Workbook